# Concept of Operations
# MinSat

TestOrg

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# Revision History

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# 1.0 Introduction

## 1.1 Purpose and Scope

This Concept of Operations (ConOps) document describes the operational concept for the MinSat mission. It provides a comprehensive description of the mission objectives, system architecture, operational modes, communication architecture, and disposal plan. This document serves as the primary operational reference for FCC licensing under Part 25.114 (Standard Application) and supports frequency coordination with the IARU and ITU.

The ConOps is prepared in accordance with ANSI/AIAA G-043A-2012 (Guide for the Preparation of Operational Concept Documents) and IEEE 1362-1998.

## 1.2 Mission Objectives

## 1.3 Applicable Documents and Standards

| Reference | Title |
|---|---|
| 47 CFR 25.114 | Applications for Space Station Authorizations |
| 47 CFR 25.122 | Streamlined Small Satellite Authorization Process |
| 47 CFR 97.207 | Space Station (Amateur Radio) |
| ANSI/AIAA G-043A-2012 | Guide for Operational Concept Documents |
| IEEE 1362-1998 | Concept of Operations Document Guide |
| CDS Rev 14.1 | CubeSat Design Specification (Cal Poly) |
| NASA-STD-8719.14 | Process for Limiting Orbital Debris |
| ITU Radio Regulations | International Frequency Coordination |

## 1.4 Regulatory Framework

This mission will operate under FCC Part 25.114 (Standard Application). The application will include all required information per 47 CFR 25.114, including orbital parameters, frequency information, system operations narrative, interference analysis, and orbital debris mitigation plan.

The FCC requires compliance with orbital debris mitigation rules, including the 5-year post-mission deorbit requirement (FCC-22-74).

# 2.0 Mission Overview

## 2.1 Mission Description and Rationale

The MinSat mission is a CubeSat designed to operate in Sun-Synchronous (SSO) at 500 km altitude. The primary payload is a Camera for . Operated by TestOrg, the mission targets a 2-year operational lifetime with full compliance to FCC regulations under Part 25.114 (Standard Application).

## 2.2 Stakeholders and Responsibilities

| Stakeholder | Role | Responsibility |
|---|---|---|
| TestOrg | Mission Owner | Overall mission management and operations |
| | Launch Services | Launch vehicle integration and deployment |
| FCC | Regulatory Authority | Spectrum licensing and debris compliance |
| | Ground Segment | Communication and command operations |

## 2.3 Mission Success Criteria

## 2.4 Mission Constraints and Assumptions

**Constraints:** Mission lifetime shall not exceed 2 years. Spacecraft mass is 4 kg. All transmissions must be capable of immediate cessation via ground command.

**Assumptions:** Launch vehicle will deliver to target orbit within specified tolerances. Ground station availability meets minimum contact requirements. Solar power generation is sufficient for all operational modes.

## 2.5 Mission Timeline

| Phase | Duration | Target Date |
|---|---|---|
| Design & Development | 12-24 months | In progress |
| Integration & Test | 3-6 months | TBD |
| FCC Filing & Coordination | 6-12 months | TBD |
| Launch Readiness Review | 1 month | TBD |
| Launch | — | |
| Nominal Operations | 2 years | L + 2 years |
| End-of-Life Disposal | < 5 years post-mission | Per FCC-22-74 |

# 3.0 Space Segment Description

## 3.1 Spacecraft Configuration

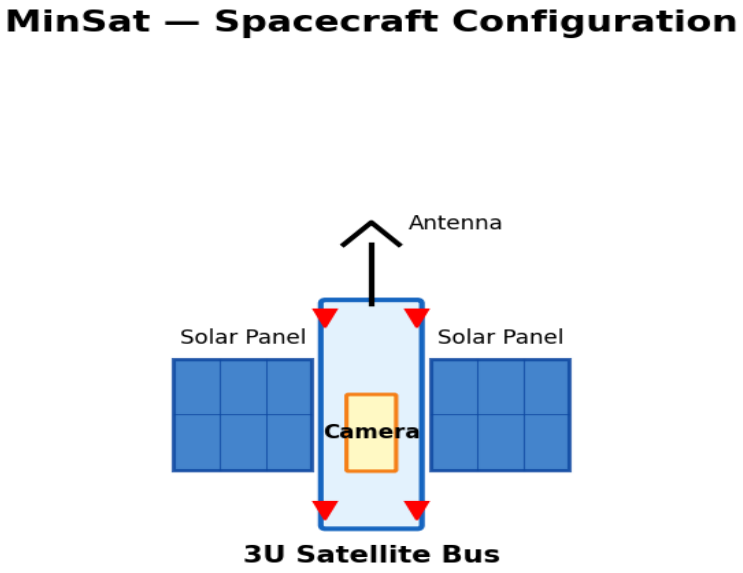


*Figure 1: MinSat spacecraft configuration diagram*

| Parameter | Value |
|---|---|
| Satellite Class | CubeSat |
| Form Factor | 3U |
| Mass | 4 kg |
| Dimensions | |
| Design Life | 2 years |
| Attitude Control | |
| Propulsion | |
| Power System | |
| On-Board Computer | |
| Unique Telemetry Marker | Embedded in downlink beacon — per 47 CFR 25.122 |

*Table 1: Spacecraft key parameters*

## 3.2 Subsystem Summary

| Subsystem | Description | Heritage / TRL |
|---|---|---|
| EPS | | TRL 7-8 |
| ADCS | | TRL 7 |
| C&DH | | TRL 8 |
| Comms | , bps | TRL 8 |
| Thermal | Passive (MLI + surface coatings) | TRL 9 |
| Propulsion | | N/A |
| Payload | Camera | TRL 5-6 |

## 3.3 Payload Description

The primary payload is a Camera designed for . Data is stored onboard and downlinked during scheduled ground station passes at bps.

# 4.0 Orbit Description

## 4.1 Orbital Parameters

| Parameter | Value |
|---|---|
| Orbit Type | Sun-Synchronous (SSO) |
| Altitude | 500 km |
| Inclination | 97.4° |
| Eccentricity | ~0 (near-circular) |
| RAAN | TBD |
| Orbital Period | 94.5 minutes |
| Orbital Velocity | 7.62 km/s |
| Orbits per Day | ~15.2 |
| LTAN | |

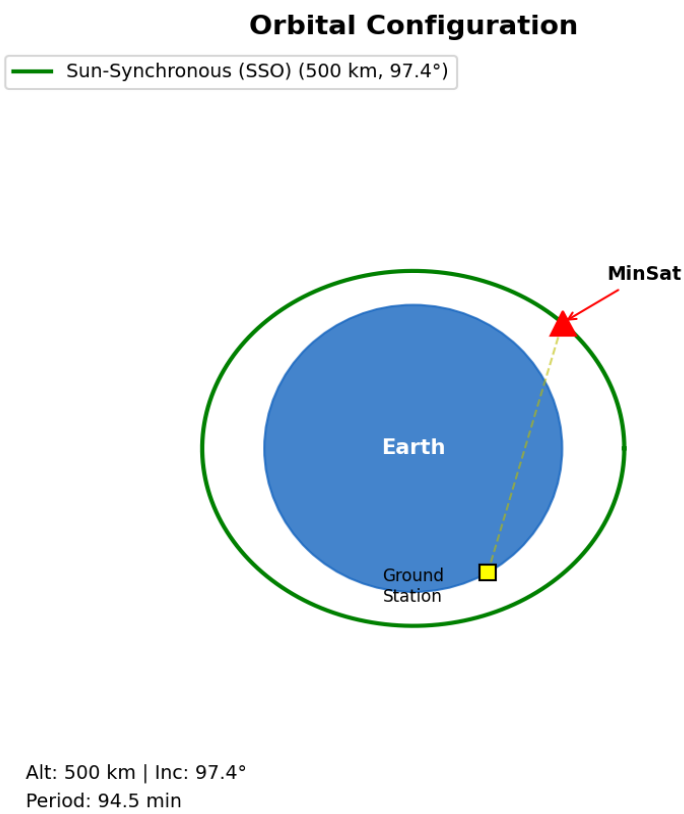


*Figure 2: Orbital configuration diagram*

## 4.2 Eclipse Analysis

At 500 km altitude, the spacecraft will experience approximately 33.1 minutes of eclipse per 94.5-minute orbit (~35% duty cycle). The power system is sized to sustain all essential subsystems through eclipse.

## 4.3 Orbital Decay Analysis

Estimated natural orbital decay time at 500 km: 3-7 years (natural decay, solar activity dependent). Active measures may be required to meet the FCC 5-year post-mission disposal requirement (FCC-22-74).

# 5.0 Mission Phases

*Figure 3: Mission phase timeline*

## 5.1 Pre-Launch (Integration & Testing)

Final spacecraft integration, environmental testing (vibration, thermal vacuum), end-to-end communication testing, and launch vehicle integration. FCC license application and IARU frequency coordination must be completed prior to launch.

## 5.2 Launch and Deployment

The spacecraft will be launched via . Post-deployment, the spacecraft will enter a pre-programmed inhibit period before activating any transmitters.

## 5.3 Early Operations / Detumbling

Upon deployment, the ADCS will execute detumbling to reduce tip-off rates. Initial beacon transmission will commence after the inhibit period. Ground station acquisition is expected within the first 24-48 hours.

## 5.4 Commissioning

Systematic checkout of all subsystems over approximately 2-4 weeks, including ADCS calibration, communication link verification, payload checkout, and flight software validation.

## 5.5 Nominal Operations

Primary mission operations lasting 2 years. The spacecraft will alternate between science data collection, data downlink, and housekeeping modes.

## 5.6 End-of-Life / Decommissioning

Upon mission completion, the spacecraft will enter disposal mode. All stored energy will be passivated. The spacecraft must deorbit within 5 years per FCC-22-74.

# 6.0 Operational Modes

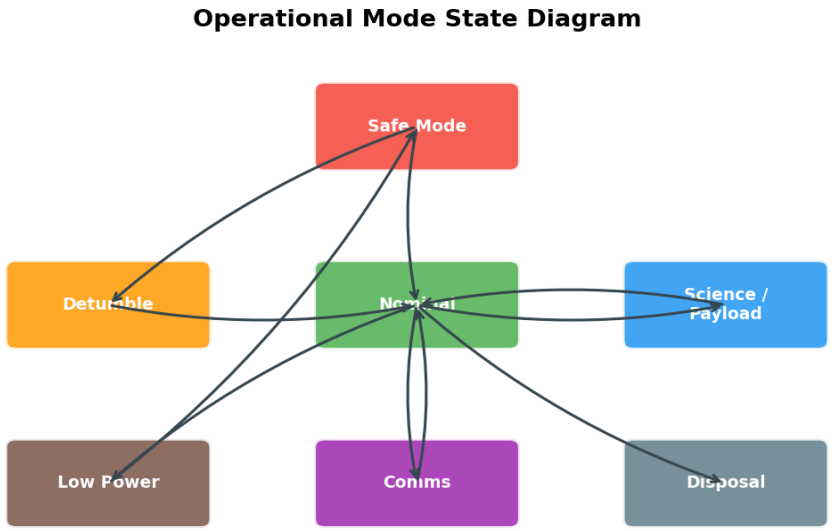


*Figure 4: Operational mode state transition diagram*

| Mode | Description | Power (W) | Duration | ADCS State |
|---|---|---|---|---|
| Safe Mode | Minimum power, beacon only, sun-pointing | Low | Until recovery | Sun-pointing |
| Detumble | Reduce angular rates post-deployment | Low-Med | Min to hours | B-dot |
| Nominal | Standard housekeeping and attitude control | Medium | Default | 3-axis |
| Science/Payload | Active payload data collection | High | Per schedule | Nadir/Target |
| Communication | Ground station pass, uplink/downlink | Med-High | ~10 min/pass | GS tracking |
| Low Power | Battery conservation during eclipse | Low | Eclipse | Coarse |
| Disposal | Deorbit maneuver / passivation | Medium | One-time | As required |

*Table 2: Operational modes summary*

## 6.1 Mode Transition Logic

Mode transitions are governed by the onboard flight software. Safe Mode is the highest priority and can be entered from any mode upon detection of an anomaly (under-voltage, communication timeout, attitude error exceedance). Ground command can override any mode transition.

# 7.0 Communication Architecture

*Per 47 CFR 25.114(d)(1) and 25.122(a), this section provides the required narrative description of the communication system operations, frequencies, and spectrum-sharing strategy.*

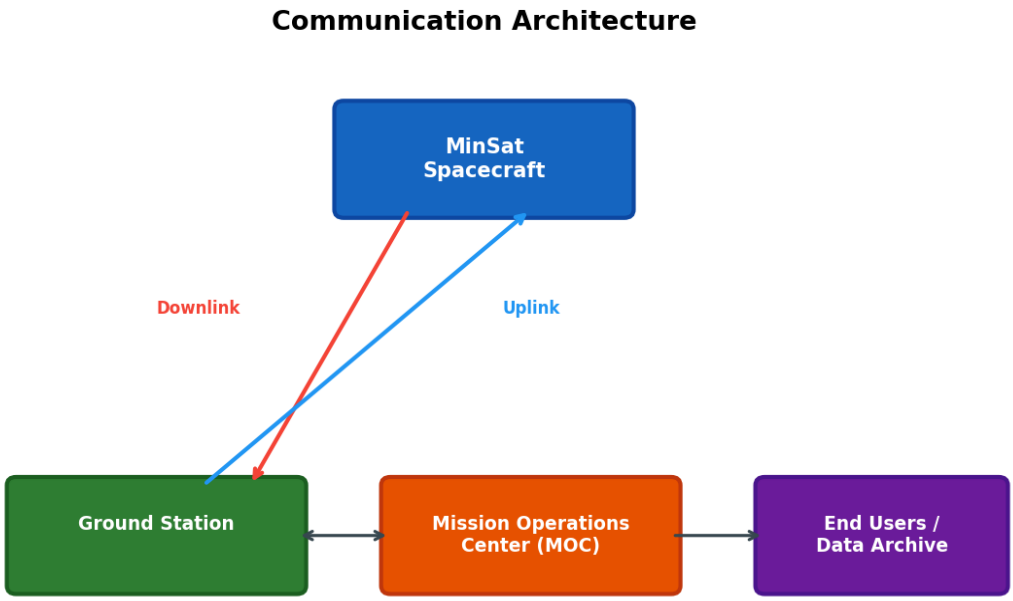


*Figure 5: Communication architecture diagram*

## 7.1 Frequency Plan

| Parameter | Uplink | Downlink |
|---|---|---|
| Frequency | | |
| Bandwidth | | |
| Modulation | GMSK | GMSK |
| Data Rate | bps | bps |
| Protocol | AX.25 | AX.25 |
| Polarization | RHCP | RHCP |
| EIRP | | — |

## 7.2 Spectrum Sharing Strategy

The MinSat mission is designed to share spectrum without constraining future satellite or terrestrial operators. The spacecraft uses narrowband transmissions with duty-cycled operation (active only during scheduled ground passes). IARU frequency coordination will be completed prior to launch.

## 7.3 Transmission Cessation Capability

Per 47 CFR 25.122(a) and 97.207, the spacecraft is capable of immediate cessation of all radio transmissions upon ground command. The telecommand link provides a dedicated 'TX OFF' command with highest priority in the command processing queue.

## 7.4 Link Budget Summary

| Parameter | Uplink | Downlink |
|---|---|---|
| Tx Power | | |
| Tx Antenna Gain | | |
| Rx Antenna Gain | | |
| System Noise Temp | 1000 K | 200 K |
| Required Eb/N0 | 10 dB | 10 dB |
| Link Margin | > 6 dB | > 6 dB |

*Table 3: Link budget summary (see Appendix A for analysis)*

# 8.0 Ground Segment

## 8.1 Ground Station Description

The primary ground station is located at . The station is equipped with a and . The ground station supports both manual and automated pass operations.

## 8.2 Contact Schedule

The ground station will have approximately 4 passes per day with an average duration of 10 minutes (total daily contact time: ~40 minutes).

## 8.3 Data Budget

| Parameter | Value |
|---|---|
| Downlink Data Rate | 9600 bps |
| Passes per Day | 4 |
| Daily Downlink Capacity | ~2.7 MB |
| Daily Payload Data Generated | |

# 9.0 Command and Data Handling

## 9.1 Command Protocol

Telecommands are transmitted via the uplink () using AX.25 protocol. Commands are authenticated to prevent unauthorized execution. Per 47 CFR 97.211, encrypted telecommand is permitted.

## 9.2 Telemetry and Health Monitoring

The spacecraft transmits periodic health beacons containing: battery voltage, solar panel current, temperatures, ADCS state, mode, and error flags. A unique signal-based telemetry marker (per 47 CFR 25.122) is embedded in every downlink frame.

## 9.3 Autonomy and Fault Management

The flight software implements a three-tier fault management architecture: (1) hardware watchdog timer, (2) software FDIR for subsystem anomalies, and (3) mission-level safe mode entry upon detection of critical faults.

# 10.0 Power Budget Analysis

This section presents the power generation and consumption analysis for each operational mode.

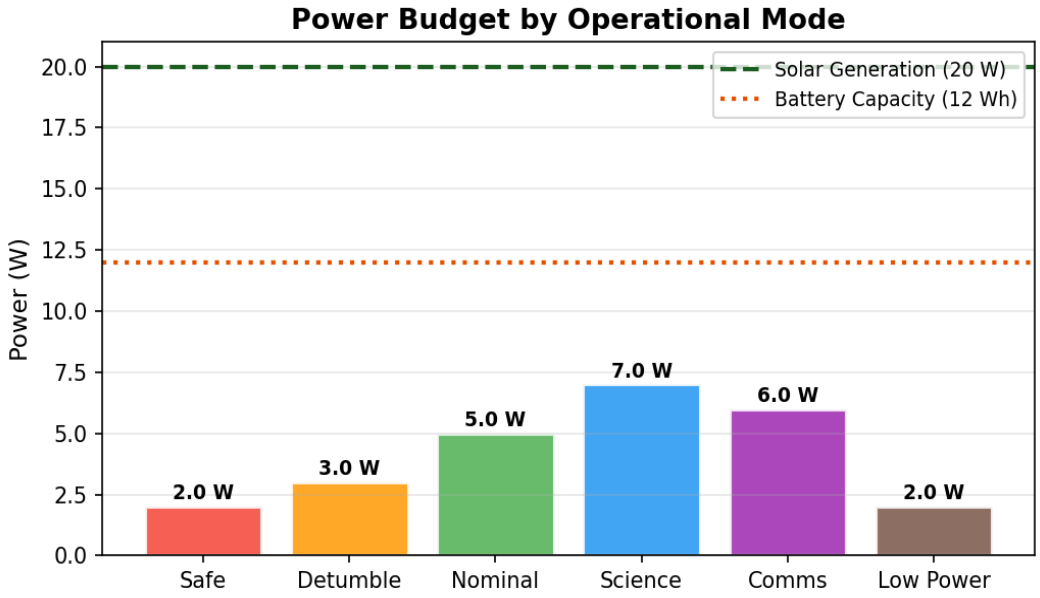


*Figure 6: Power budget by operational mode*

| Parameter | Value |
|---|---|
| Peak Solar Generation | 20 W |
| Battery Capacity | 12 Wh |
| Battery Type | Li-ion |
| Solar Panel Area | |
| Eclipse Duration (per orbit) | 33.1 min |

# 11.0 Orbital Debris Mitigation Plan

*Per 47 CFR 25.114(d)(14) and FCC-22-74 (5-year deorbit rule).*

## 11.1 Operational Debris Release

No debris will be released during normal mission operations. All deployment mechanisms use captive fasteners.

## 11.2 Collision Risk Assessment

| Requirement | Threshold | Assessment | Compliance |
|---|---|---|---|
| Small debris collision (<10 cm) | P < 0.01 | TBD (NASA DAS) | Expected compliant |
| Large object collision (>10 cm) | P < 0.001 | TBD (NASA DAS) | Expected compliant |
| Accidental explosion | Minimize probability | Passivation at EOL | Compliant |

## 11.3 Trackability

The MinSat spacecraft meets minimum dimension requirements for tracking by the 18th Space Defense Squadron Space Surveillance Network. The spacecraft will be registered with the U.S. Space Command catalog.

## 11.4 Conjunction Assessment and Response

The mission operations team will subscribe to conjunction data messages (CDMs) from the 18th Space Defense Squadron. Ephemeris data will be shared with the Space Surveillance Network to support conjunction screening.

# 12.0 End-of-Life Disposal

*Per 47 CFR 25.114(d)(14) and FCC-22-74 (5-year deorbit rule).*

## 12.1 Deorbit Method

Disposal method: Atmospheric reentry (uncontrolled) — natural drag.

| Parameter | Value |
|---|---|
| Disposal Method | Atmospheric reentry (uncontrolled) — natural drag |
| Post-Mission Orbital Lifetime | < 5 years (per FCC-22-74) |
| Disposal Success Probability | ≥ 0.9 (per 47 CFR 25.122) |
| Human Casualty Risk | < 1:10,000 (per 47 CFR 25.114) |

## 12.2 Passivation

At end of life, all stored energy will be removed: batteries discharged, pressurized systems vented, reaction wheels de-spun. This eliminates accidental explosion risk per 47 CFR 25.114(d)(14)(ii).

## 12.3 Casualty Risk Assessment

For a 4 kg satellite, complete demise during atmospheric reentry is expected, resulting in zero casualty risk.

# 13.0 Risk Assessment and Contingency Operations

## 13.1 Mission Risk Register

| ID | Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|---|
| R1 | Launch delay | Medium | Medium | Flexible manifest, rideshare options |
| R2 | Communication loss | Low | High | Redundant GS, autonomous safe mode |
| R3 | ADCS failure | Low | High | Backup modes, sun-pointing safe mode |
| R4 | Power degradation | Medium | Medium | Oversized arrays, battery management |
| R5 | Payload malfunction | Low | Medium | Pre-flight testing, calibration |
| R6 | Debris collision | Very Low | Very High | Conjunction monitoring, avoidance |
| R7 | Software anomaly | Medium | Medium | Watchdog, safe mode, patching |
| R8 | Deorbit failure | Low | High | Drag augmentation backup |

*Table 5: Mission risk register*

## 13.2 Contingency Operations

**Loss of Communication:** If no ground contact within 72 hours, spacecraft will autonomously cycle through communication modes. After 30 days, permanent safe mode.

**Safe Mode Recovery:** Entered upon under-voltage, attitude error, or comm timeout. Recovery requires ground command after root cause analysis.

# 14.0 Regulatory Compliance Summary

| Requirement | Regulation | Status | Section |
|---|---|---|---|
| FCC Space Station License | Part 25.114 (Standard Application) | In preparation | §1.4 |
| Orbital Parameters | 47 CFR 25.114(c) | Provided | §4.0 |
| System Operations Narrative | 47 CFR 25.114(d)(1) | Provided | §5.0–§9.0 |
| Communication Architecture | 47 CFR 25.114(c)(4) | Provided | §7.0 |
| Spectrum Sharing Strategy | 47 CFR 25.122(a) | Provided | §7.2 |
| Transmission Cessation | 47 CFR 25.122(a) / 97.207 | Implemented | §7.3 |
| Unique Telemetry Marker | 47 CFR 25.122(a) | Designed | §3.1 |
| Debris Mitigation Plan | 47 CFR 25.114(d)(14) | Provided | §11.0 |
| Collision Risk Assessment | 47 CFR 25.114(d)(14)(iii) | Pending | §11.2 |
| Trackability | 47 CFR 25.114(d)(14)(v) | Compliant | §11.3 |
| Conjunction Response | 47 CFR 25.114(d)(14)(vi) | Planned | §11.4 |
| 5-Year Deorbit | FCC-22-74 | Compliant | §12.0 |
| Casualty Risk < 1:10,000 | 47 CFR 25.114(d)(14)(iv) | Assessed | §12.3 |
| Passivation Plan | 47 CFR 25.114(d)(14)(ii) | Planned | §12.2 |
| IARU Frequency Coordination | IARU Form V40 | In preparation | §7.1 |

*Table 6: FCC regulatory compliance cross-reference matrix*

# 15.0 References


• 47 CFR Part 25 — Satellite Communications (https://www.law.cornell.edu/cfr/text/47/part-25)

• 47 CFR 25.114 — Applications for Space Station Authorizations (https://www.law.cornell.edu/cfr/text/47/25.114)

• 47 CFR 25.122 — Streamlined Small Satellite Process (https://www.law.cornell.edu/cfr/text/47/25.122)

• 47 CFR Part 97 — Amateur Radio Service (https://www.law.cornell.edu/cfr/text/47/part-97)

• 47 CFR 97.207 — Space Station (https://www.law.cornell.edu/cfr/text/47/97.207)

• ANSI/AIAA G-043A-2012: Guide for the Preparation of Operational Concept Documents

• IEEE 1362-1998: IEEE Guide for Information Technology — System Definition — Concept of Operations Document

• CubeSat Design Specification Rev 14.1 (https://www.cubesat.org/cubesatinfo)

• NASA CubeSat Launch Initiative Resources (https://www.nasa.gov/content/cubesat-launch-initiative-resources)

• NASA GSFC-STD-1001A: Flight Project Lifecycle Reviews (https://standards.nasa.gov/standard/GSFC/GSFC-STD-1001)

• IARU Satellite Frequency Coordination (https://www.iaru.org/reference/satellites/)

• FCC 5-Year Deorbit Rule, FCC-22-74 (2022)

• ITU Radio Regulations (https://www.itu.int/pub/R-REG-RR)

# Appendix A: Link Budget Analysis

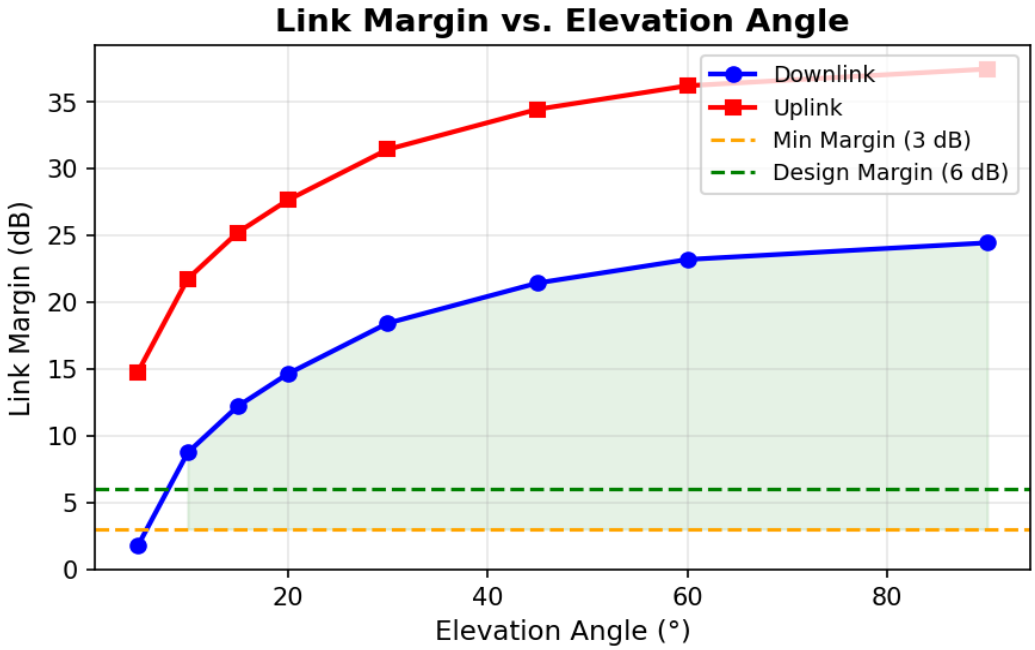


*Figure 7: Link margin vs. elevation angle*

| Parameter | 5° Elevation (Worst) | 90° Elevation (Best) |
|---|---|---|
| Slant Range | 5737 km | 500 km |
| Free Space Path Loss | 160.4 dB | 139.2 dB |
| Atmospheric Loss | 2.0 dB | 0.5 dB |
| Total Path Loss | 163.4 dB | 140.2 dB |

# Appendix B: FCC Application Cross-Reference Matrix

| FCC Requirement | CFR Citation | ConOps Section |
|---|---|---|
| Applicant Information | 25.114(a) | Title Page |
| Orbital Parameters | 25.114(c)(1)-(5) | §4.0 |
| Frequency Information | 25.114(c)(4) | §7.1 |
| Antenna Characteristics | 25.114(c)(4)(iii) | §7.4 |
| System Operations | 25.114(d)(1) | §5.0–§9.0 |
| Interference Analysis | 25.114(d)(3) | §7.2 |
| Cessation of Emissions | 25.122(a) | §7.3 |
| Debris Mitigation | 25.114(d)(14) | §11.0 |
| Collision Avoidance | 25.114(d)(14)(iii) | §11.2, §11.4 |
| End-of-Life Disposal | 25.114(d)(14)(iv) | §12.0 |
| Trackability | 25.114(d)(14)(v) | §11.3 |
| Casualty Risk | 25.114(d)(14)(iv) | §12.3 |
| Unique Telemetry Marker | 25.122(a) | §3.1 |
| Maneuverability | 25.122(a) | §3.1 |

# Appendix C: Acronyms and Abbreviations

| Acronym | Definition |
|---|---|
| ADCS | Attitude Determination and Control System |
| C&DH | Command and Data Handling |
| CDM | Conjunction Data Message |
| ConOps | Concept of Operations |
| DAS | Debris Assessment Software |
| EIRP | Effective Isotropic Radiated Power |
| EOL | End of Life |
| EPS | Electrical Power System |
| FCC | Federal Communications Commission |
| FDIR | Fault Detection, Isolation, and Recovery |
| GEO | Geostationary Earth Orbit |
| IARU | International Amateur Radio Union |
| ITU | International Telecommunication Union |
| LEO | Low Earth Orbit |
| LTAN | Local Time of Ascending Node |
| MEO | Medium Earth Orbit |
| MLI | Multi-Layer Insulation |
| MOC | Mission Operations Center |
| NGSO | Non-Geostationary Orbit |
| RAAN | Right Ascension of the Ascending Node |
| RHCP | Right-Hand Circular Polarization |
| SDS | Space Defense Squadron |
| SSO | Sun-Synchronous Orbit |
| TRL | Technology Readiness Level |